# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-SIDEREAL-EN · v2.2 · 2026-11-02 |
| Package | star-sidereal |
| Version | 0.1.2 |
| Capability | star.time.sidereal |
| Maturity | Released & Verified (RELEASED) |
| Python | ⩾3.9 |
| Source commit | dc09f61 |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_sidereal |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_sidereal-0.1.2-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_sidereal-0.1.2-py3-none-any.whl | 81cebed335b95316594d92c82d9246c4c20b328ea10a898ab84f410da66f715d |
| star_sidereal.py | adb187e78d0b9f2a7a7150f3cb385611596c71d7ce9325e7ff526dd8c94a9d3e |
| star_sidereal-0.1.2-source.zip | b21f4add552a067f420f43431e4be3c32ca3e4aad3f02da0d5000709fca6786e |

# 5. What it does, and its limits

GMST and local mean sidereal time, IAU 1982 (Vallado eq. 3-47); verified against Vallado and ERFA.

IAU 2000/2006 (Earth rotation angle based) sidereal time, apparent sidereal time (equation of the equinoxes), UTC→UT1.

- `gmst82_deg(jd_ut1_day, jd_ut1_frac=0.0)` → degrees in [0, 360), Vallado eq. 3-47 (Aoki et al. 1982).
- `lst_deg(jd_day, jd_frac, east_longitude_deg)` → degrees in [0, 360).
- Non-finite input raises `ValueError`. Input is UT1 (not UTC): the caller applies DUT1.

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

# 7. Citation and bill of materials

```
@software{cavazzini_star_sidereal_0_1_2,
  author  = {Cavazzini, Andrea},
  title   = {{star-sidereal}: Greenwich mean sidereal time (IAU 1982)},
  version = {0.1.2},
```

```
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_sidereal/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts